Week 6 — Activity: Curie Depth and Magnetic Crustal Thickness

Geophysics 3A: Potential Fields & Geothermics

April 10, 2026

Purpose

This activity explores how temperature limits crustal magnetisation and why deep magnetic structure is long-wavelength.

Part A: Curie Depth Concept

1. What mineral dominates crustal magnetisation?
2. What happens above the Curie temperature?
3. Is Curie depth a thermal or compositional boundary?

Part B: Depth and Wavelength

Given several magnetic profiles:

1. Which correspond to shallow sources?
2. Which to deep Curie depth?
3. How does depth affect short wavelengths?

Part C: Antarctica Case Study

Consider the magnetic and crustal structure of Antarctica (Fig. 1).
Compare:

- Magnetic anomalies
- Seismic crustal thickness
- Magnetic crustal thickness

Why do seismic and magnetic thickness estimates differ?

The first-order geology of Antarctica is characterized by a stable (cratonic) region in East Antarctica and a Cenozoic rift in West Antarctica (Fig. 2).

Part D: Heat-Flow Estimate

Assume:

$$T_C = 570°\text{C}, \quad k = 2.8\ \text{W m}^{-1}\text{K}^{-1}$$

$$q = k\frac{T_C}{z}$$

Estimate heat flow for a rift and a craton.

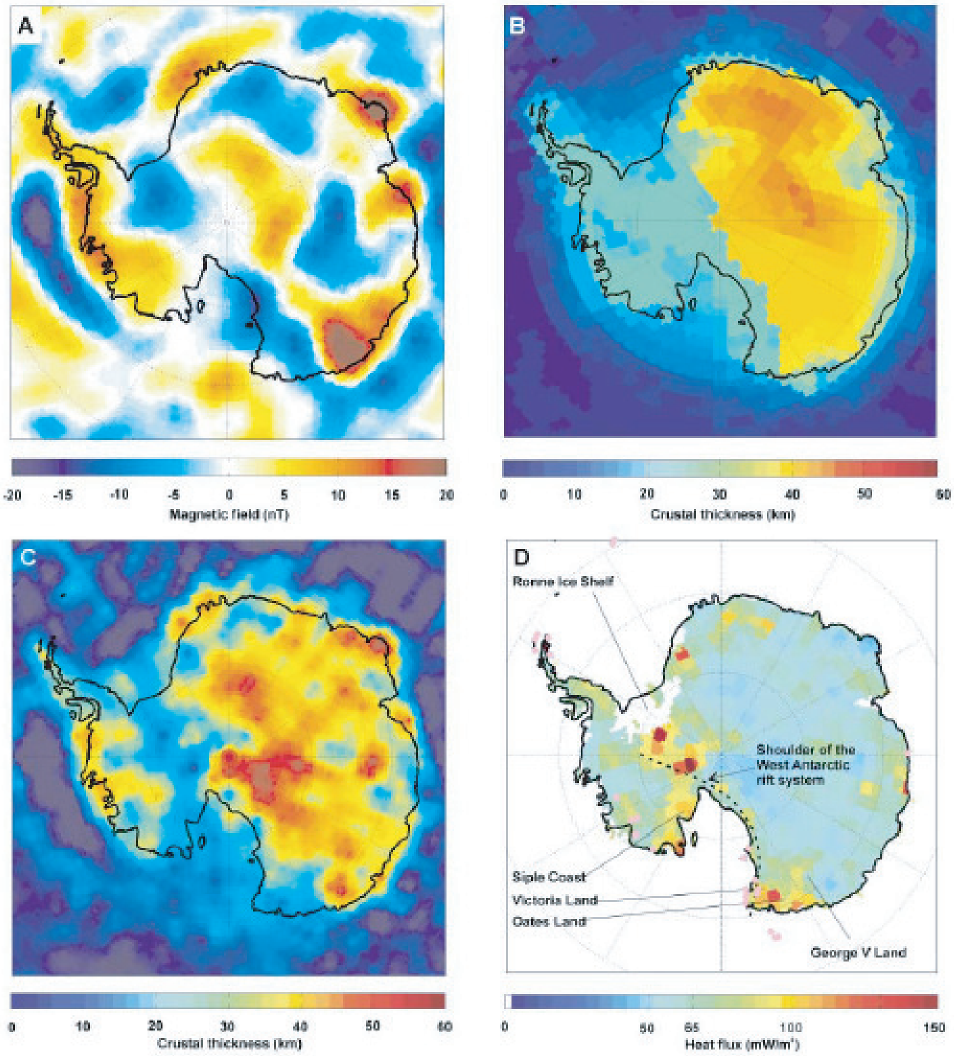
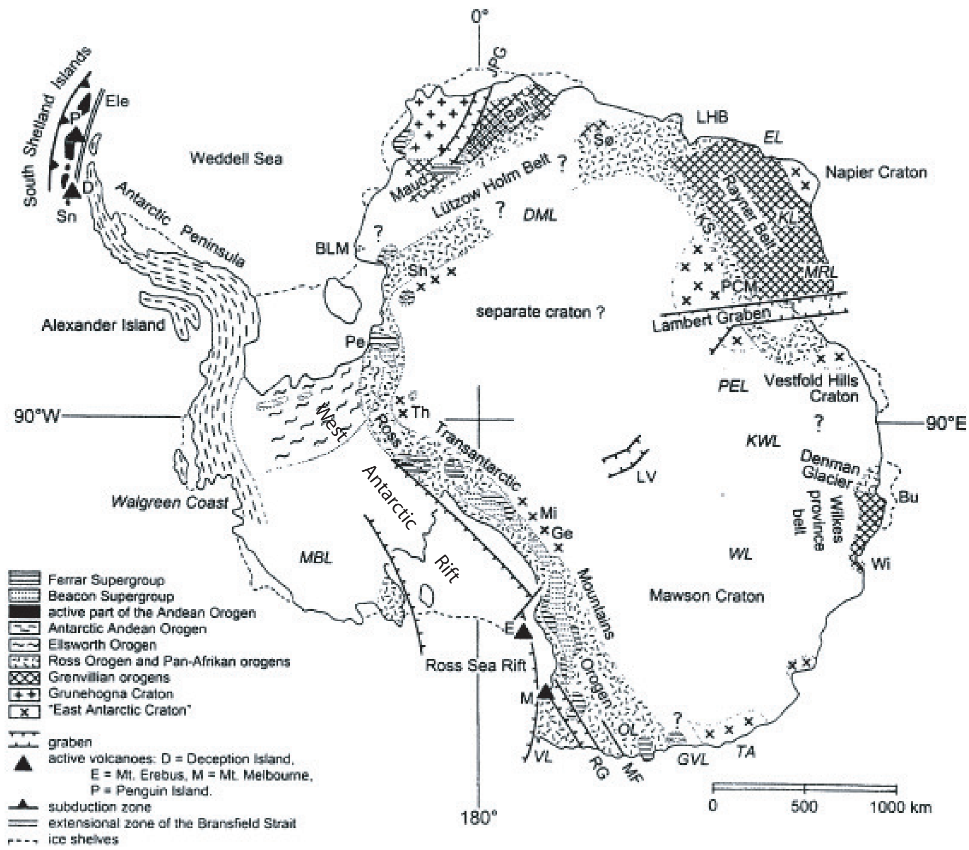


Figure 1: Estimating the Curie depth from vertical magnetic data. (a) Vertical magnetic field. (b) Crustal thickness model of Antarctica. (c) Estimate to base of magnetized layer. (d) Estimated heat flow. Figure from Maule et al. (**Science**, 2005).

Reflection

Explain why Curie depth makes magnetics sensitive to thermal structure.



Talarico and Kleinschmidt, in **Antarctic Climate Evolution**, 2008

Figure 2: Tectonic map of Antarctica from Talarico and Kleinschmidt (**Antarctic Climate Evolution**, 2008).